Supplementary Materials

Seizure Forecasting and Crash Risk Analysis

June 3, 2026

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1 Overview

This appendix gives the full model, the parameter derivations, the main result, the sensitivity analyses, and the reproducibility instructions for the main paper. All calculations are analytic (closed-form); the accompanying Jupyter notebook `crashes_vs_TiW.ipynb` reproduces every figure and every number in this document.

We quantify the trade-off between seizure-forecasting accuracy (AUC), driving exposure, and the driving restrictions required to maintain fatal crash risk at or below a safety threshold of approximately 6× baseline, corresponding to BAC ≈ 0.05–0.07% for adult drivers. This benchmark aligns with the 0.05% BAC standard used in Utah and most European countries, more conservative than the 0.08% BAC limit in 49 U.S. states.

Key modeling feature. Relative fatal crash risk on forecasted "safe" days is computed using a per-hour fatal-crash baseline $\lambda_f \approx 4.5 \times 10^{-7}\,\mathrm{hr}^{-1}$ and a conditional probability $q_f = P(\text{fatal crash} \mid \text{seizure while driving}) = 0.02$ (central estimate; range 0.005–0.05 in sensitivity analyses). The probability that a seizure coincides with driving scales as $D/24$ for daily driving duration D; however, because both seizure-related and baseline risk scale with exposure, the relative-risk benchmark remains approximately invariant to D (Section 2.4).

Key findings. Patients with infrequent seizures (≤ 1/year) approach unrestricted driving even at low AUC, whereas patients with frequent seizures require AUC ≥ 0.90 to drive meaningfully. At AUC 0.90 and 1 hr/day driving, weekly seizures permit ≈ 53 days/year, monthly seizures ≈ 182, and yearly seizures ≈ 365. These results depend little on driving duration but moderately on q_f (Section 4).

2 Model and Methods

This section builds the model in the order it is computed: forecasting model, seizure prior, Bayesian posterior on forecast-safe days, fatal-crash-risk model, safety threshold, and conversion to driving days.

2.1 Binormal ROC Framework

We model the forecasting algorithm's output as a scalar score S generated by one of two normal distributions, depending on whether the day is a seizure day.

Variable definitions used throughout this appendix.

- S: the forecasting algorithm's output score on a given day.
- m: the separation between the seizure and non-seizure score distributions (larger m means the algorithm discriminates better).
- t: the operating threshold; the algorithm warns *unsafe to drive* when $S > t$ and forecasts *safe to drive* when $S \leq t$.
- $\Phi(\cdot)$: the cumulative distribution function (CDF) of the standard normal distribution $N(0,1)$.
- $\Phi^{-1}(\cdot)$: the inverse standard-normal CDF (quantile function).
- AUC: area under the receiver operating characteristic curve.

The forecast score is distributed as

$$S \sim N(0,1) \quad \text{on a non-seizure day,} \qquad S \sim N(m,1) \quad \text{on a seizure day.} \tag{2.1}$$

Under this equal-variance binormal model, the separation m relates to AUC by

$$m = \sqrt{2}\,\Phi^{-1}(\text{AUC}). \tag{2.2}$$

For threshold t applied to the score, the sensitivity (true-positive rate) and false-positive rate are

$$\text{Sensitivity} = P(S > t \mid \text{seizure}) = 1 - \Phi(t - m) = \Phi(m - t), \tag{2.3}$$

$$\text{FPR} = P(S > t \mid \text{no seizure}) = 1 - \Phi(t). \tag{2.4}$$

Figure S1 illustrates the model geometrically: the two score distributions, the threshold t, the separation m, and the regions whose probability mass equals sensitivity (2.3) and FPR (2.4). Section 4.3 tests robustness to alternative ROC shapes.

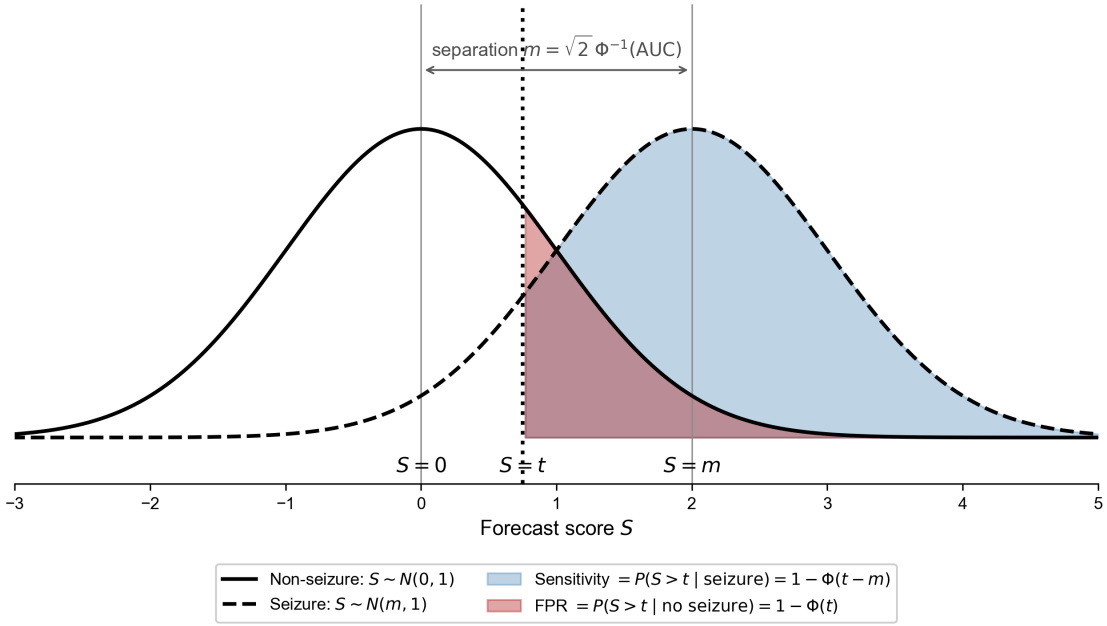


Figure S1: **Binormal forecast model.** Score distributions on non-seizure days ($S \sim N(0, 1)$, solid) and seizure days ($S \sim N(m, 1)$, dashed). The threshold t (dotted vertical) defines the decision boundary; sensitivity (blue) and FPR (red) are the probability masses right of t under the seizure and non-seizure distributions. Separation parameter m relates to AUC via Equation 2.2.

2.2 Daily Seizure Probability

For a patient with average seizure rate R (seizures per day), the daily seizure probability under a Poisson process is

$$p = 1 - e^{-R}. \tag{2.5}$$

Frequency	R	p
1 seizure / week	1/7	≈ 0.133
1 seizure / month	1/30	≈ 0.033
1 seizure / year	1/365	≈ 0.0027

The Poisson mapping is used only to translate an average rate into a daily probability; the analysis is otherwise agnostic to the temporal structure of seizure risk.

2.3 Posterior Seizure Probability on Forecast-Safe Days

When a patient drives on a forecast-safe day (one for which the algorithm output $S \leq t$), the residual crash risk is driven by the posterior probability of a seizure given that negative forecast. By Bayes' theorem,

$$P(\text{seizure} \mid \text{forecast safe}) \;=\; \frac{P(\text{forecast safe} \mid \text{seizure}) \cdot p}{P(\text{forecast safe} \mid \text{seizure}) \cdot p + P(\text{forecast safe} \mid \text{no seizure}) \cdot (1-p)}. \tag{2.6}$$

We obtain the two conditional probabilities in (2.6) directly from sensitivity and FPR by complementation:

$$P(\text{forecast safe} \mid \text{seizure}) \;=\; 1 - P(\text{forecast unsafe} \mid \text{seizure}) \;=\; 1 - \text{Sensitivity}, \tag{2.7}$$

$$P(\text{forecast safe} \mid \text{no seizure}) \;=\; 1 - P(\text{forecast unsafe} \mid \text{no seizure}) \;=\; 1 - \text{FPR}. \tag{2.8}$$

Substituting (2.7)–(2.8) into (2.6) and using the binormal expressions for sensitivity (2.3) and FPR (2.4) gives

$$P(\text{seizure} \mid \text{forecast safe}) \;=\; \frac{(1-\text{Sensitivity})\,p}{(1-\text{Sensitivity})\,p + (1-\text{FPR})\,(1-p)} \;=\; \frac{\Phi(t-m)\,p}{\Phi(t-m)\,p + \Phi(t)\,(1-p)}. \tag{2.9}$$

2.4 Fatal Crash Risk and the Relative-Risk Derivation

Parameters used in this subsection. We use the parameters in Table 1 throughout the crash-risk derivation.

Table 1: Crash-risk model parameters.

Symbol	Meaning	Value	Source
λ_f	Baseline fatal-crash rate per hour of sober driving	$4.5 \times 10^{-7}\,\text{hr}^{-1}$	NHTSA *Passenger Vehicles 2023* + FHWA *Highway Statistics 2023* + AAA *American Driving Survey 2023* (derivation in this section)
q_f	P(fatal crash | seizure while driving)	0.02 central; range 0.005–0.05	Inferred severity parameter; see Section 4.1
D	Driving hours per day	0.5, 1, 2	—
π	P(seizure | forecast safe)	From Eq. (2.9)	—

Baseline per-day fatal-crash risk. Given D hours of driving per day and a per-hour baseline rate λ_f, a sober driver faces a per-day baseline fatal-crash risk of

$$B(D) \;=\; D \cdot \lambda_f. \tag{2.10}$$

Derivation of λ_f. NHTSA *Passenger Vehicles 2023* reports $\sim$ 44,000 passenger vehicles involved in fatal crashes; FHWA *Highway Statistics 2023* Table VM-1 reports $\sim$ 2.88 trillion light-duty vehicle-miles travelled (VMT). The ratio is approximately 1.5 passenger-vehicle fatal-crash involvements per 10^8 miles, i.e. 1.5×10^{-8} per mile. Converting per-mile to per-hour requires a driving speed: the AAA *American Driving Survey 2023* reports ≈ 29.1 miles over ≈ 60.7 minutes per driver per day, i.e. ≈ 29 mph. Then

$$\lambda_f \;\approx\; 1.5 \times 10^{-8}\,\text{mile}^{-1} \times 29\,\text{mph} \;\approx\; 4.4 \times 10^{-7}\,\text{hr}^{-1} \;\approx\; 4.5 \times 10^{-7}\,\text{hr}^{-1}. \tag{2.11}$$

(FHWA Table VM-1 supplies the VMT denominator only; the AAA survey provides driving speed for the per-mile-to-per-hour conversion.) Because this baseline incorporates all non-seizure causes of fatal crashes (distraction, weather, other drivers), the model conservatively attributes this background risk to every driving day regardless of seizure status.

Seizure-day fatal-crash risk. On a day when a seizure occurs, the per-day fatal-crash risk depends on whether the seizure falls within the driving window. Assuming uniform seizure timing across the 24-hour day,

$$P(\text{fatal crash} \mid \text{seizure day}) \;=\; \frac{D}{24} \cdot q_f \;+\; \left(1 - \frac{D}{24}\right) \cdot B(D). \tag{2.12}$$

The first term captures seizures during driving; the second captures baseline risk when the seizure occurs outside driving hours.

Forecast-safe-day fatal-crash risk. Let $\pi = P(\text{seizure} \mid \text{forecast safe})$ from Equation 2.9. On a forecast-safe day, residual fatal-crash risk is

$$P(\text{fatal crash} \mid \text{forecast safe}) \;=\; \pi \cdot P(\text{fatal crash} \mid \text{seizure day}) \;+\; (1-\pi) \cdot B(D). \tag{2.13}$$

Relative fatal-crash risk and the exposure-invariance derivation. Substituting (2.12) into (2.13) and dividing by the baseline $B(D)$ defines the relative risk:

$$\text{RR}(D) \;=\; \frac{P(\text{fatal crash} \mid \text{forecast safe})}{B(D)} \;=\; \frac{\pi\left[\dfrac{D}{24}\, q_f + \left(1 - \dfrac{D}{24}\right) B(D)\right] + (1-\pi)\, B(D)}{B(D)}. \tag{2.14}$$

Distribute the numerator and divide each term by $B(D) = D\lambda_f$:

$$\text{RR}(D) \;=\; \pi \cdot \frac{(D/24)\, q_f}{D\, \lambda_f} \;+\; \pi\left(1 - \frac{D}{24}\right) \;+\; (1-\pi). \tag{2.15}$$

The D in the first term cancels:

$$\text{RR}(D) \;=\; \pi \cdot \frac{q_f}{24\, \lambda_f} \;+\; \pi \;-\; \pi \cdot \frac{D}{24} \;+\; 1 - \pi. \tag{2.16}$$

The π and $-\pi$ cancel, leaving the exact expression

$$\boxed{\mathrm{RR}(D) \;=\; 1 + \pi\left[\frac{q_f}{24\,\lambda_f} - \frac{D}{24}\right]}. \tag{2.17}$$

Under our parameters, $q_f/(24\lambda_f) = 0.02/(24 \times 4.5 \times 10^{-7}) \approx 1852$, while $D/24 \le 2/24 \approx 0.083$ for $D \le 2$ hours. Since $1852 \gg 0.083$, the $D/24$ term is negligible and we approximate

$$\mathrm{RR}(D) \;\approx\; 1 + \pi \cdot \frac{q_f}{24\,\lambda_f} \;=\; 1 + \pi \cdot A, \qquad A \;\equiv\; \frac{q_f}{24\,\lambda_f}. \tag{2.18}$$

A is the *amplification factor*.

Exposure invariance. The approximation (2.18) makes the D-dependence vanish: at $D = 1$ hr, the residual $D/24$ correction reaches at most $0.083\,\pi$ versus the dominant $\sim 1852\,\pi$ (a relative discrepancy of order 5×10^{-5}). Absolute fatal-crash risk scales with driving exposure through $B(D)$, but the *relative* risk against the BAC-anchored threshold remains approximately invariant to D. Section 4.2 confirms this invariance numerically.

2.5 Safety Threshold

We use a safety threshold derived from the BAC 0.05% standard. Under the doubling-per-0.02-BAC model of Zador et al. and Compton & Berning, the relative fatal-crash risk at BAC 0.05% is

$$T \;=\; 2^{\,0.05/0.02} \;\approx\; 5.66, \tag{2.19}$$

approximately 6× baseline. Throughout this appendix we call this a "6× threshold" for readability; the code uses the precise value $T = 2^{0.05/0.02}$. Given this threshold, we find the operating threshold t such that

$$\frac{P(\text{fatal crash} \mid \text{forecast safe})}{B(D)} \;\le\; T, \tag{2.20}$$

and report the corresponding minimum days in warning (the minimum restriction required to meet the threshold).

2.6 Time in Warning and Driving Days

The proportion of days the algorithm flags as high-risk (the "time in warning") is

$$P(\text{warning}) \;=\; \text{Sensitivity} \cdot p + \text{FPR} \cdot (1 - p), \tag{2.21}$$

$$\text{Days in warning per year} \;=\; P(\text{warning}) \times 365. \tag{2.22}$$

The complement, $(1 - P(\text{warning})) \times 365$, gives days per year the patient is forecast safe to drive. An algorithm that warns every day achieves perfect sensitivity but permits zero driving days; for each AUC, the analysis therefore finds the operating threshold t that just meets the safety constraint (2.20) and reports the resulting driving days.

Table 2: Driving restrictions required to achieve the safety threshold by seizure frequency and forecasting performance (1 hr/day driving).

Seizure freq.	AUC	Warning days/yr	Driving days/yr	Avg. days between drives
1/week	0.60	365	0	Cannot reach safety threshold
	0.70	365	0	Cannot reach safety threshold
	0.80	363	2	~ 237
	0.90	312	53	6.9
	0.95	221	144	2.5
	0.99	91	274	1.3
1/month	0.60	365	0	Cannot reach safety threshold
	0.70	364	1	~ 497
	0.80	323	42	8.8
	0.90	183	182	2.0
	0.95	92	273	1.3
	0.99	23	342	1.1
1/year	0.60	34	331	1.1
	0.70	8	357	1.0
	0.80	2	363	1.0
	0.90	0	365	1.0 (already safe)
	0.95	0	365	1.0 (already safe)
	0.99	0	365	1.0 (already safe)

3 Main Result

Table 2 reports the driving restrictions required to achieve the safety threshold by seizure frequency and forecasting performance, at 1 hr/day driving, $q_f = 0.02$, $\lambda_f = 4.5 \times 10^{-7}\,\text{hr}^{-1}$.

Figure S2 shows the underlying mechanism: how crash risk on forecast-safe days falls as the algorithm spends more days in warning, and where each AUC curve crosses the safety threshold.

Key insights.

- *High AUC is the binding constraint.* Algorithms need AUC ≥ 0.90 to provide meaningful driving days for patients with weekly or monthly seizures.
- *Patients with rare seizures ($\leq$ 1/year)* are at or near unrestricted driving even at low AUC ($\approx$ 331 days/year at AUC 0.60), though not literally always-safe.
- *Monthly seizure patients benefit greatly*: AUC 0.90 permits $\approx$ 182 days/year; AUC 0.95 permits $\approx$ 273.
- *Weekly seizure patients face significant restrictions*: even AUC 0.90 permits only $\approx$ 53 days/year; AUC 0.80 is essentially insufficient ($\leq$ 2 days/year).

4 Sensitivity Analyses

Three analyses test the robustness of the main result: to the most uncertain parameter (q_f, Section 4.1), to driving duration (Section 4.2), and to ROC-curve shape (Section 4.3).

4.1 Conditional Fatal-Crash Probability q_f

The parameter $q_f = P(\text{fatal crash} \mid \text{seizure while driving})$ is the model's most uncertain component; no single study directly reports it, making it an inferred severity parameter. We examined the defensible range $q_f \in [0.005,\ 0.05]$:

- **Lower bound ($q_f = 0.005$):** moderate crash rate (~ 0.30) given a seizure while driving $\times$ moderate case-fatality (~ 0.02) — closer to baseline fatal-crash severity.
- **Central ($q_f = 0.02$):** Gastaut & Zifkin 1987 observed crash rate (~ 0.55) $\times$ case-fatality of seizure-related crashes ($\sim$ 3–5%, inferred from Sheth et al. 2004 and Drazkowski et al. 2003).
- **Upper bound ($q_f = 0.05$):** counts near-misses as effective crashes (Gastaut: $\sim 38\%$ additional near-miss rate) with upper-bound severity.

Table 3 reports driving days/year at AUC = 0.90, 1 hr/day, across the q_f range.

Table 3: Sensitivity of driving days/year to q_f at AUC = 0.90, 1 hr/day.

Seizure frequency	$q_f = 0.005$	$q_f = 0.02$ (central)	$q_f = 0.05$
1/week	154	53	20
1/month	314	182	99
1/year	365	365	334

The amplification factor $A = q_f/(24\lambda_f)$ ranges from ≈ 463 (lower) through ≈ 1852 (central) to ≈ 4630 (upper); the corresponding maximum tolerable $P(\text{seizure} \mid \text{forecast safe})$ for RR $\leq T = 5.66$ ranges from $\approx 1.0\%$ down to $\approx 0.10\%$.

Interpretation. Across the full range, the qualitative finding holds: weekly-seizure patients cannot achieve unrestricted driving even at AUC 0.90, and yearly-seizure patients remain at or near unrestricted driving. The quantitative allowance for monthly seizures spans roughly a factor of three. Until empirical data better anchor the parameter, stakeholders may prefer the conservative-end $q_f = 0.05$.

4.2 Driving Duration (Exposure Invariance)

Section 2.4 shows analytically that relative risk is approximately invariant to driving duration D, because both seizure-related and baseline fatal risks scale with exposure (Equation 2.18). This invariance also holds numerically: driving-day allowances coincide to integer rounding across the examined durations (Table 4).

Table 4: Driving days/year coincide across exposure levels under the relative-risk benchmark.

Seizure frequency	AUC	30 min/day	1 hr/day	2 hr/day
1/week	0.90	53	53	53
1/month	0.90	182	182	182
1/year	0.80	363	363	363

Absolute fatal-crash risk does still scale with driving duration; an alternative *absolute*-risk benchmark would depend directly on exposure. Under the relative-risk benchmark used here, driving duration is not a material determinant of the allowable driving days.

4.3 ROC Curve Shape

The equal-variance binormal assumption implies a specific sensitivity–specificity relationship. Real ROC curves can deviate from this relationship due to unequal variances (S-shaped), mixture distributions (hooked; some seizures easier to predict than others), or finite-sample effects. We test three score-generating models, all calibrated to AUC ≈ 0.90.

Equal-variance binormal (baseline): negatives follow $N(0,1)$ and positives follow $N(m,1)$, with $m = \sqrt{2}\,\Phi^{-1}(0.90) \approx 1.81$. Equations (2.3)–(2.4) give sensitivity and FPR.

Unequal-variance binormal (S-shaped): negatives $N(0,1)$, positives $N(\mu_1, \sigma_1)$ with $\mu_1 = 2.4$, $\sigma_1 = 1.4$. For threshold t,

$$\text{Sensitivity} \;=\; 1 - \Phi\left(\frac{t-\mu_1}{\sigma_1}\right), \qquad \text{FPR} \;=\; 1 - \Phi(t). \tag{4.1}$$

On forecast-safe days, the posterior substitutes $\Phi((t-\mu_1)/\sigma_1)$ for $\Phi(t-m)$ in Equation 2.9.

Mixture model (hooked): negatives $N(0,1)$, positives $w\,N(\mu_{\text{easy}},1) + (1-w)\,N(\mu_{\text{hard}},1)$, with $w = 0.7$, $\mu_{\text{easy}} = 2.5$, $\mu_{\text{hard}} = 0.8$. From this mixture,

$$\text{Sensitivity} \;=\; w\,[1-\Phi(t-\mu_{\text{easy}})] \;+\; (1-w)\,[1-\Phi(t-\mu_{\text{hard}})], \tag{4.2}$$

$$P(S < t \mid \text{seizure}) \;=\; w\,\Phi(t-\mu_{\text{easy}}) \;+\; (1-w)\,\Phi(t-\mu_{\text{hard}}). \tag{4.3}$$

Table 5 reports driving days at AUC ≈ 0.90 and 1 hr/day for each ROC shape; Figure S5 shows the corresponding crash-risk-versus-warning-days curves across shapes and seizure frequencies.

Table 5: Driving days/year at AUC ≈ 0.90, 1 hr/day, by ROC shape.

Seizure frequency	Equal-variance binormal	S-shaped (unequal var.)	Hooked (mixture)
1/week	53	Cannot reach safety	1
1/month	182	145	66
1/year	365	365	365

Findings.

- *At high seizure frequencies, ROC shape matters.* Equal-variance binormal is the most optimistic shape; the S-shaped form fails to reach the threshold at all for weekly seizures, and the hooked mixture model permits only 1 day. For monthly seizures, the spread is 182 → 145 → 66 days/year.

- *At low seizure frequency (yearly), shape is irrelevant* — all three ROC families permit essentially full-year driving because the prior seizure probability is already very low.
- *Net.* AUC remains the primary determinant, but ROC geometry is a meaningful secondary factor. Because equal-variance binormal is the most optimistic of the three, real-world (S-shaped or hooked) ROCs at the same AUC will generally permit *fewer* driving days than the main result suggests — so reporting AUC alone may overstate clinical utility for patients with frequent seizures. The qualitative conclusion (high AUC is necessary, and even AUC 0.90 is borderline for weekly seizures) is robust to shape.

5 Assumptions, Limitations, and Recommendations

Why the binormal model is a reasonable base case. It is a standard approximation in diagnostic testing and signal-detection theory, is mathematically tractable (closed-form, reproducible), and for a given AUC tends to offer near-best-case sensitivity-specificity trade-offs — giving the forecasting algorithm the benefit of the doubt. The paper's question is "what level of performance (AUC) is needed for safe driving decisions?", not "how does one specific empirical ROC curve perform?". Section 4.3 quantifies the effect of departures from this assumption.

Driving-exposure assumptions.

1. **Seizure timing is uniformly distributed** over the 24-hour day. Many patients have circadian patterns (nocturnal epilepsy, morning clustering); the model is conservative for seizures concentrated outside driving hours and optimistic for daytime clustering.
2. **Driving duration is constant across days.** Real driving varies; the 30 min/day–2 hr/day range brackets typical commuter and heavy-driver patterns (and, per Section 4.2, barely affects the relative-risk result).
3. $q_f = 0.02$ **is an inferred severity parameter** (range 0.005–0.05), decomposing as $P(\text{crash} \mid \text{seizure while driving}) \approx 0.55$ (Gastaut & Zifkin 1987) $\times$ case-fatality ≈ 3–5% (inferred from Sheth et al. 2004 and Drazkowski et al. 2003). It depends on seizure type, aura, road conditions, and vehicle safety features; Section 4.1 reports the sensitivity analysis.

Recommendations for algorithm developers.

1. Aim for AUC ≥ 0.90 as a minimum for driving applications.
2. Characterise your empirical ROC shape — an S-shaped or hooked curve at the same AUC permits fewer driving days and shifts optimal operating points.
3. For patients with frequent seizures, even AUC $= 0.90$ may be insufficient for practical driving policies.
4. For patients with rare seizures, forecasting may be unnecessary for safety; focus on other quality-of-life benefits.
5. Consider patient-specific seizure frequency and driving patterns when counselling about risk.

6 Reproducibility

All calculations use closed-form expressions implemented with NumPy, SciPy (`scipy.stats.norm` for the Gaussian CDF/inverse CDF), and Matplotlib.

```
pip install numpy matplotlib scipy
jupyter notebook crashes_vs_TiW.ipynb        # interactive
# or run all cells programmatically:
jupyter nbconvert --to notebook --execute crashes_vs_TiW.ipynb
```

Table 6: Notebook cell layout and outputs.

Cell	Produces
0	Figure S2 — crash risk vs. days in warning (3 frequencies, 1 hr/day)
1	Figure 1 — driving days vs. AUC (main result)
2	Figure S1 — binormal model schematic
3	Figures S4 & S5 — ROC shape geometries and sensitivity
4–5	Extended AUC table (tab-delimited, all frequencies/durations)
6	Figure S3 — q_f sensitivity

Figure files are produced as PDF and PNG: `Figure_1`, `Figure_S1`–`Figure_S5`.

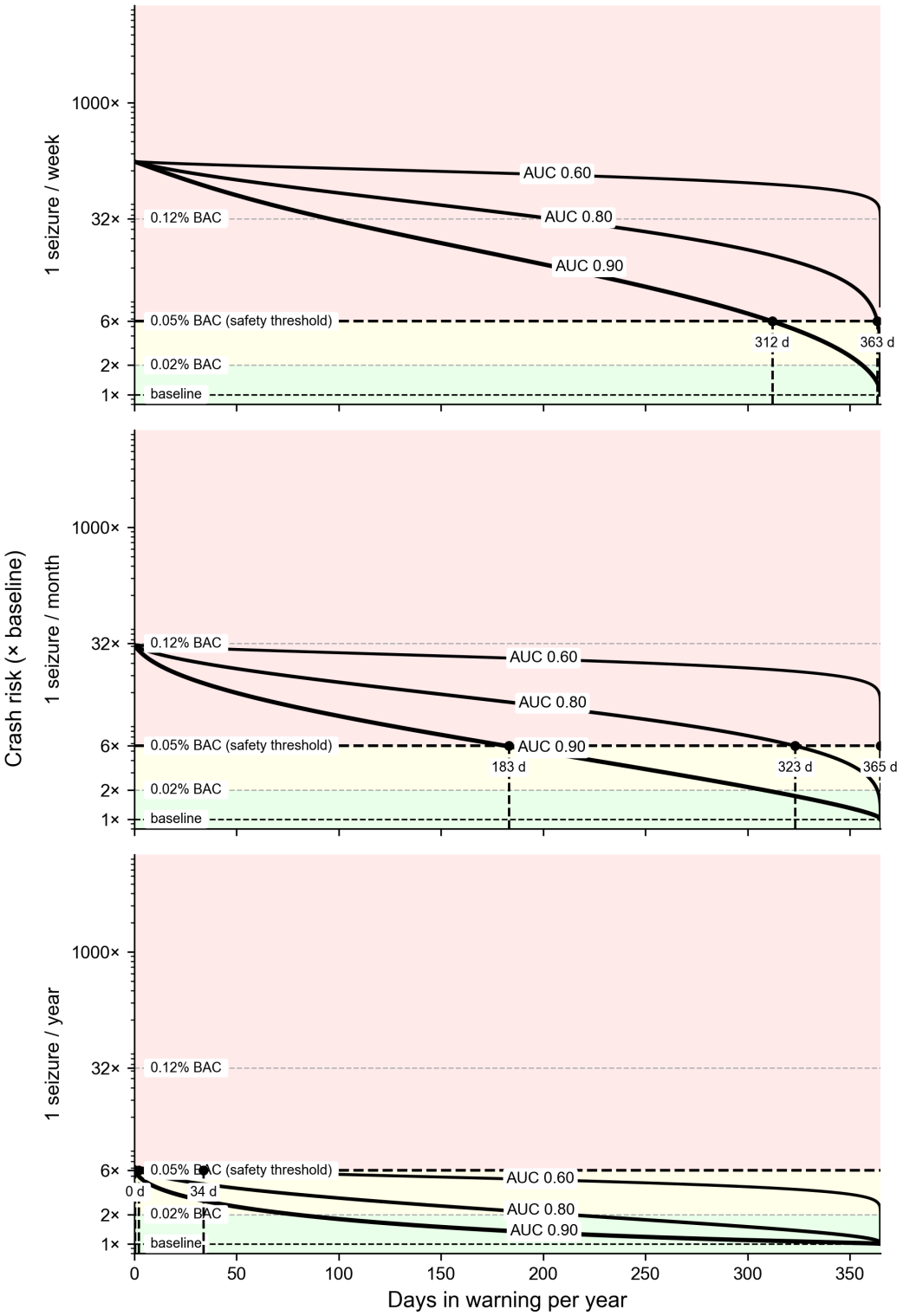


Figure S2: **Mechanism: crash risk versus days in warning.** Rows are seizure frequencies (1/week, 1/month, 1/year), at 1 hr/day driving. Black curves are AUC 0.60, 0.80, 0.90 (thicker lines indicate higher AUC). Shaded bands and dashed lines mark BAC-equivalent risk levels (0.02%, the 0.05% safety threshold, 0.12%). Black circles mark the minimum days in warning needed to reach the threshold.

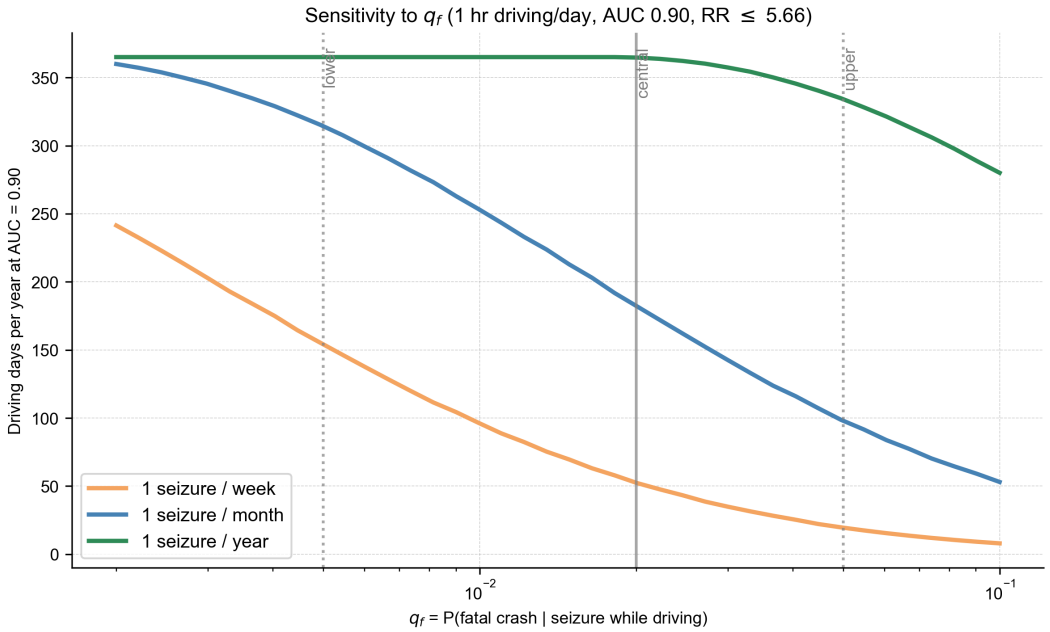


Figure S3: **Sensitivity of driving days to q_f.** Driving days per year at AUC = 0.90 and 1 hr/day, as a function of q_f. Vertical guides mark the lower bound (0.005), the central estimate (0.02), and the upper bound (0.05).

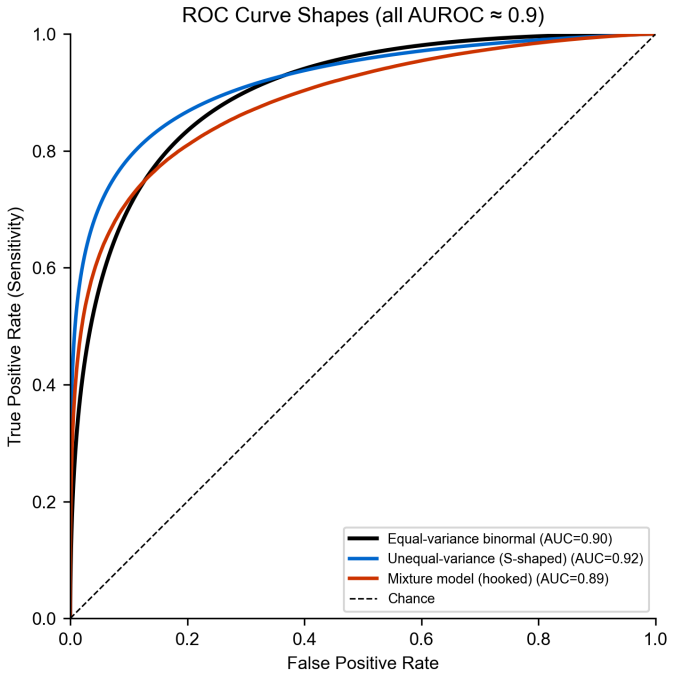


Figure S4: **Three ROC curve shapes**, all at AUC ≈ 0.90. Equal-variance binormal (black, central), unequal-variance/S-shaped (blue, top-left), and hooked mixture model (red).

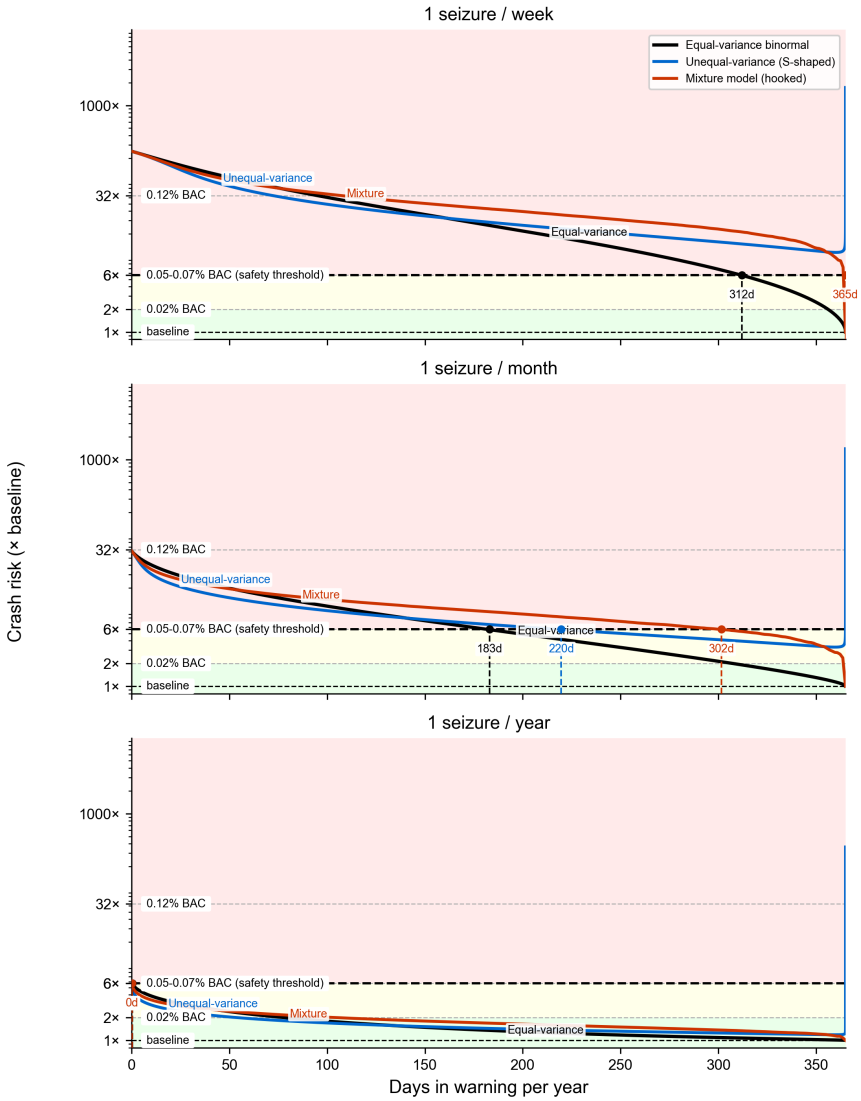


Figure S5: **Crash risk versus days in warning for three ROC shapes**, by seizure frequency, at 1 hr/day and AUC ≈ 0.90. Equal-variance binormal (black), S-shaped (blue), and hooked mixture (red). Black dashed line: safety threshold.